

A Calibrated Posterior for Wireless Capsule Endoscope Localization at a Four-Receiver MICS/MedRadio Budget: A Full-Wave Simulation Study

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Abstract—This is a full-wave electromagnetic CST Studio Suite (CST) simulation study; we report no measured-phantom or in-vivo results. We study wireless capsule endoscope (WCE) localization at a four-receiver budget relevant to the Medical Implant Communication Service (MICS)/MedRadio band (401 MHz to 406 MHz). We ask whether such a system can report calibrated confidence in capsule position, not only a point estimate. Full-wave channels are simulated over a 380 MHz to 430 MHz sweep; strict-legal-band evidence is the single-tone magnitude-only ablation. On a corrected cuboid phantom geometry (receivers on the $\pm x/\pm z$ vertical walls, no y -face coverage) and a strictly position-grouped 5-fold cross-validation protocol, we benchmark a conditional flow-matching (CFM) posterior against a deterministic regression backbone, probabilistic baselines (deep ensemble, MC-dropout, mixture density networks, split conformal), and ten classical regressors. Beyond point accuracy, the CFM posterior is calibrated: its marginal 50/90/95% coverages track the nominal levels (0.766/0.935/0.944 on the held-out initial-stage subset, $N = 1980$; 0.832/0.970/0.975 when all combined-set positions are scored by the CFM alone), and its predicted spread ranks actual error (Spearman $\rho \approx 0.60$). Point accuracy is within 0.014 cm of the deterministic backbone (0.354 cm vs 0.340 cm 3D MAE on the combined $N = 3935$ set). A controlled mirror-symmetric array ablation further indicates, on this controlled geometry, that posterior modality tracks forward-operator invertibility: the CFM posterior is bimodal when the ablation array makes y unobservable and unimodal on the main anti-symmetric side-wall array. The same posterior width carries the orientation-induced uncertainty measured across five axis-aligned capsule poses ($\rho \approx 0.85$). A feature-deployability ablation locates the single-tone, magnitude-only floor at ~ 1.5 cm and shows that the sub-centimetre regime requires phase-coherent receivers. We make no claim to state-of-the-art point accuracy: at the four-receiver MICS/MedRadio budget, real-phantom systems are more accurate, and we declare the simulation-vs-real and frequency-band confounds explicitly. To our knowledge, this is the first flow-matching/generative WCE-RF position posterior

with empirical multi-level coverage under position-grouped cross-validation.

Index Terms—Wireless capsule endoscopy, RF localization, uncertainty quantification, calibration, conformal prediction, conditional flow matching, generative posterior, MICS/MedRadio band, simulation-based inference.

I. INTRODUCTION

Wireless capsule endoscopy (WCE) lets a swallowable camera image the entire gastrointestinal (GI) tract without sedation, but a finding—a bleed, an ulcer, a polyp—is only clinically actionable if the clinician knows *where* along the tract it occurred. Radio-frequency (RF) localization of the capsule from a small set of body-surface receivers is the leading non-imaging route to that coordinate. The overwhelming majority of WCE-RF localization methods, however, report only a *point* estimate of capsule position. For a quantity a clinician must act on, a point estimate without a calibrated notion of confidence is incomplete: a 2 cm estimate that is reliably within 2 cm and a 2 cm estimate that is silently off by 6 cm in the unobservable direction are clinically very different, yet a point estimator cannot tell them apart.

At a four-receiver budget for the legal Medical Implant Communication Service (MICS)/MedRadio band 401 MHz to 406 MHz, the scientifically and clinically valuable contribution is the calibrated posterior: a distribution over capsule position whose spread is an interpretable, usable measure of uncertainty. We study the four-receiver MICS/MedRadio budget with the main coherent sub-band at ~ 402 MHz to 405 MHz [1], and ask whether a generative model can deliver such a posterior *without* sacrificing point accuracy.

Scope (declared up front): All results in this paper come from full-wave CST Studio Suite (CST) simulation; no measured-phantom or in-vivo data are used. Simulation-first evaluation is standard practice in this subfield: comparable studies report on simulation or simulation-dominant evidence [2], [3], [4], [5], [6], [7]. We use full-wave CST simulation (not a parametric path-loss model), evaluate under a protocol *stricter* than the random point-wise splits these comparators use, and identify a physical-phantom companion as the explicit next evidence gate (Section VI). Figure 1 summarizes this scope and shows that the evidence chain

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This is a full-wave electromagnetic (CST) simulation study; no measured-phantom or in-vivo results are reported.

targets calibrated posterior reporting, not a point-accuracy leaderboard.

Contributions and scope: **(A) What we do not claim.** We are not the most accurate point estimator at the four-receiver MICS/MedRadio budget. Real-phantom received signal strength indicator (RSSI) systems at 2.45 GHz report better point accuracy [8], [9], our results are simulation at a $\sim 6\times$ lower frequency, and our protocol is stricter; we do not contest the point-accuracy leaderboard. **(B) What we do claim: the contribution is the calibrated posterior.** To our knowledge, this is the first flow-matching/generative WCE-RF position posterior with empirical multi-level coverage under position-grouped cross-validation. Concretely:

- 1) A clean conditional flow-matching (CFM) posterior that is calibrated *from its own samples*: near-nominal multi-level coverage (0.766/0.935/0.944 on the initial-stage subset; 0.832/0.970/0.975 in combined-set CFM-only scoring) and real error-spread discrimination (Spearman $\rho \approx 0.60$).
- 2) Point accuracy within 0.014 cm of the deterministic backbone (0.354 cm vs 0.340 cm 3D MAE) while significantly beating ten classical regressors on identical position-grouped folds (position-paired bootstrap, Section V/Table V).
- 3) The empirical observation that posterior *modality* tracks forward-operator invertibility in a controlled receiver-array ablation: the CFM posterior is bimodal when a mirror-symmetric array leaves y unobservable by construction, and unimodal on the main anti-symmetric side-wall array where the receiver placement breaks that symmetry.
- 4) Evidence that the same posterior width carries the aleatoric uncertainty of an *unknown* capsule orientation, reframing orientation as a nuisance the posterior integrates over rather than a quantity to be regressed.
- 5) A feature-deployability ablation on the initial-stage subset ($N = 1980$) that separates the single-tone, magnitude-only floor (~ 1.5 cm) from the phase-coherent sub-centimetre regime; these ablation values are therefore not the combined-set $N = 3935$ headline numbers.

The one Bayesian WCE-RF precedent [5] reports RMSE against the Cramér–Rao lower bound on a parametric path-loss simulation with no empirical coverage, which is why we frame the novelty around the learned posterior and its empirical coverage rather than around Bayesian localization in general. Section II surveys WCE-RF localization and the calibrated-posterior gap; Section III describes the simulation and feature pipeline; Section IV defines the CFM posterior; Section V reports point accuracy, coverage, modality, deployability, and orientation uncertainty; and Section VI and Section VII state the scope and physical-phantom gate.

II. RELATED WORK

This section separates three axes: point-estimator reporting versus calibrated posterior reporting, probabilistic precedent, and simulation-first scope relative to the subfield norm.

A. WCE RF localization is dominated by point estimators

The closest-regime WCE-RF systems are mostly point estimators or filtered point tracks. Real-phantom RSSI work at 2.45 GHz with four wearable receivers reports 2.38 cm RMSE [9] and, more recently, 1.69 cm 3D RMSE [8]; lower-frequency four-receiver work at 433 MHz reports 1.9 cm in simulation and 2.22 cm on an electromagnetic phantom [2]. Temporal filtering does appear here: the Anzai/Ito lineage evaluates RSSI and hybrid TOA/RSSI particle-filter tracking [10], [11], [12], [13], and Dmitrieva *et al.* evaluate constant-velocity interacting-multiple-model extended-Kalman filtering (IMM-EKF) for WCE tracking [14]. These tracking systems still report point-track errors or error CDFs, not empirically calibrated position posteriors or multi-level coverage.

The two closest Hasnain phantom studies report “tracking” as path RMSE between measured path positions and per-snapshot ML predictions. The PDFs do not document a Kalman, particle-filter, or Rauch–Tung–Striebel (RTS) smoother. Table I situates our work against the audited landscape. The comparison spans heterogeneous regimes—simulation vs phantom vs in-vivo, MICS/MedRadio vs ISM vs UWB, differing receiver counts and evaluation protocols—so we declare those confounds wherever numbers appear and make no like-for-like ranking.

B. Bayesian and bound-based prior art

The closest probabilistic precedent is Nadimi *et al.* [5], which combines a Rayleigh-fading, Gamma path-loss-exponent likelihood with Gaussian anchor priors and reports ~ 1.6 mm RMSE over Monte-Carlo trials with five anchors. That work is *parametric path-loss simulation* and reports RMSE and one-sigma bounds against the Cramér–Rao lower bound (CRLB), but not empirical multi-level coverage. Bound-based analyses [4], [22], [18] similarly characterize achievable accuracy analytically rather than delivering a calibrated estimator. Our positioning is therefore deliberately narrow and verifiable: not “first Bayesian” and not “first temporal filter,” but, to our knowledge, the first flow-matching/generative WCE-RF position posterior with empirical multi-level coverage under position-grouped cross-validation, including posterior *modality*—a behaviour the single-mode parametric likelihoods used in this subfield cannot represent.

C. Generative posteriors and simulation-based inference

Treating localization as posterior inference over an intractable forward operator places this work in the simulation-based-inference (SBI) and conditional generative-model literature. Conditional flow matching [23], [24] learns a continuous-time transport from noise to a data distribution conditioned on observations, and conditional generative regressors such as CARD [25] target the full predictive distribution rather than a point. We adopt the CFM formulation as our CFM posterior; the architecture itself is not the novelty, the *application and the calibration evidence* are.

Calibrated posterior pipeline for RF capsule localization

full-wave simulation evidence now; physical phantom validation is the next gate

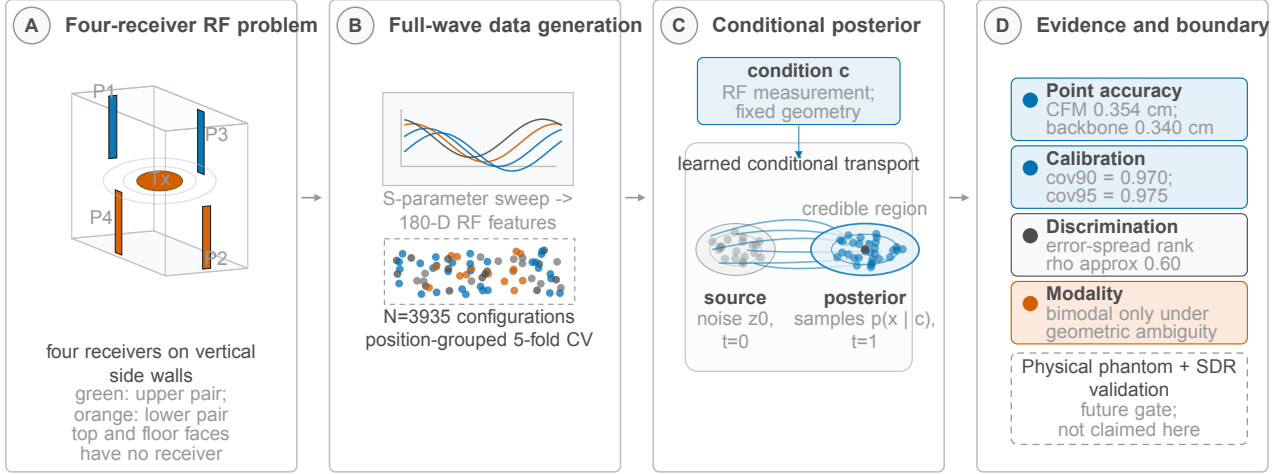


Fig. 1. Study overview. A four-receiver MICS/MedRadio budget WCE localization problem is evaluated with full-wave CST simulation, converted to rich RF features under position-grouped cross-validation, and solved with a conditional flow-matching posterior. The figure supports the paper’s main claim: the contribution is the calibrated posterior, with multi-level coverage and error-spread discrimination while point accuracy remains within 0.014 cm of the backbone (0.354 cm vs 0.340 cm). **Simulation only:** the physical phantom setup is the next validation gate, not a result claimed in this paper.

D. Calibration and conformal prediction

Distribution-free calibration via (split) conformal prediction [26], [27] provides finite-sample coverage targets under exchangeability and is the natural baseline against which to judge a learned posterior. We include split conformal as a baseline; the question we ask of a generative posterior is whether it is calibrated *and* discriminative (its width ranks error) without such a post-hoc set wrapper.

E. Simulation-first evaluation is standard in this subfield

Simulation-first evaluation is the norm in this subfield: nine of the audited systems [10], [12], [13], [14], [6], [7], [5], [4], [21] are simulation or simulation-dominant, published in venues from EMBC to IEEE Access, IEICE, and PLoS ONE. Physical-phantom and in-vivo work [20], [2], [15], [17], [9], [8] sets the bar for the companion validation we identify as future work.

III. GEOMETRY, SIMULATION, AND DATASET

A. Receiver geometry

Our four-receiver array covers only the vertical side walls—the $\pm x$ and $\pm z$ faces—with no coverage on the $\pm y$ top/floor faces, which makes y the hardest axis. An anti-symmetric height placement (P1, P3 at $y = +6.8$ cm; P2, P4 at $y = -7.3$ cm) breaks the $y \rightarrow -y$ mirror symmetry enough to recover y in simulation. The phantom is a benchtop tank with internal dimensions 273 (x) $\times$ 278 (y) $\times$ 380 (z) mm, where y is the vertical (height) axis. Only the separate mirror-symmetric ablation array in Section V-D is unobservable by construction. Geometry exploration is closed for this study; the asymmetry is treated as a fixed property of a four-receiver rig, not a free parameter. Figure 2 confirms that the $\pm y$ faces carry no

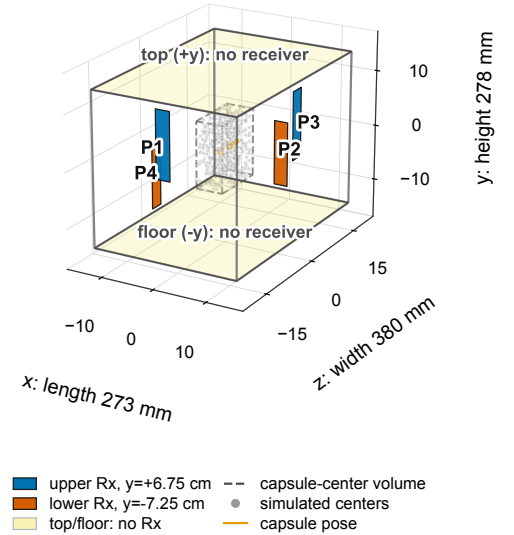


Fig. 2. Four-receiver side-wall array used by the processed simulation dataset, drawn with y (height) as the vertical axis. The outer wireframe is the 273 (x) $\times$ 278 (y) $\times$ 380 (z) mm tank envelope; the shaded top (+ y) and floor (− y) faces carry no receiver, confirming that y has no face coverage. The dashed box marks the capsule-center sampling volume; the grey cloud shows the 788 simulated capsule centres. The four receiver patch symbols sit on the vertical $\pm x/\pm z$ side walls and are placed anti-symmetrically in height (green: P1/P3 upper, $y = +6.8$ cm; orange: P2/P4 lower, $y = -7.3$ cm), breaking the $y \rightarrow -y$ symmetry; contrast the unobservable mirror-symmetric ablation in Section V-D (Fig. 15). **Simulation.**

receiver coverage and that the main array uses anti-symmetric receiver heights.

TABLE I
REPRESENTATIVE WCE-RF LOCALIZATION LANDSCAPE. **OUR WORK IS FULL-WAVE CST SIMULATION.** REGIMES AND METRICS ARE HETEROGENEOUS; THIS TABLE IS FOR POSITIONING, NOT A LIKE-FOR-LIKE RANKING. THE TAKEAWAY IS THAT NONE REPORT EMPIRICALLY CALIBRATED MULTI-LEVEL COVERAGE (FINAL COLUMN).

Work	Evidence	Frequency	#Rx	Protocol	Metric	Reported value	Regime verdict
This work	Full-wave sim (CST)	380–430 MHz full-wave sweep; 401–406 MHz MICS/MedRadio	4 ($\pm x / \pm z$)	Pos.-grouped 5-fold	3D MAE	0.354 cm rich; ~1.5 cm mag-only	internal simulation reference; calibrated coverage; not point-SOTA
Anzai 2012 [10]	Mixed sim	400 MHz MICS	14 (8 ideal)	MC FDTD + path-loss	3D RMS	4 cm ML (M=3); LS 5 cm	low-frequency RSSI baseline; no posterior
Ito/Anzai 2013–2016 [11], [12], [13]	Sim tracking	400 MHz MICS; UWB 3.4–4.8 GHz	8	MC / SMC trajectory	RMS track error	0.7 cm RSSI PF; 1.3 mm hybrid TOA/RSSI PF	temporal-filter point-track prior art
Dmitrieva 2021 [14]	Sim tracking	not stated (RSSI model)	8	single simulated track	RMSE	~8 mm constant-velocity IMM-EKF	temporal-filter point-track prior art
Pourhomayoun 2014 [6]	Sim (Monte Carlo)	406 MHz MICS	4–16	500 MC runs	RMS error	<7 mm 3D / <8.8 mm 2D	MICS sparse-recovery point estimate
Nadimi 2014 [5]	Parametric path-loss sim	not stated	5	1000 MC, RMSE/CRLB	RMSE	1.6 mm	Bayesian parametric prior; no empirical coverage
Hiyoshi 2017 [7]	Sim (FDTD)	433.92 MHz	5	792-point grid	pass-rate threshold	<40 mm over full grid; $\geq 80\%$ within 20% threshold	low-frequency RSS correction; no mean/RMSE
Yoshitake 2022 [2]	Sim + EM phantom	433.92 MHz	4	fixed 78+78 points	mean 3D error	19 mm sim / 22.2 mm phantom	close low-frequency comparator
Oleksy 2022/2023 [15], [16]	Sim + phantom / sim	401–406 MHz MICS	4–5 (select 5/9)	fixed positions, CDF/mean	3D / 2D error	16 mm sim; 30 mm 2D phantom; 42 mm sim (2023)	MICS PDoA, adaptive body model; no posterior
Xiao 2022 [17]	Mixed sim + phantom	433 MHz	24–32	MC sim + 5 phantom pts	3D RMSE	2.5 mm sim; 36.3 mm phantom	low-frequency WCL; many receivers
Hany 2023 [18]	Parametric path-loss sim	UWB 3.1–6 GHz	8–48	fixed trajectory + CRLB	3D RMSE	6.83 mm @48Rx; 8.7 mm @8Rx	analytical/path-loss context; no posterior
Narmatha 2022 [19]	Hybrid RF+vision sim	UWB path-loss model	8–80	MATLAB time-step eval.	ALE / RMSE	5.41 mm ALE; 6.76 mm RMSE	hybrid vision+RF, not RF-only
Barbi 2019 [20]	Phantom + in-vivo	UWB 3.1–5.1 GHz	best 3/5; best 10/13	measured subset selection	2D / in-vivo error	0.72 cm 2D phantom; 0.94 cm in-vivo	physical-validation reference
Hasnain 2024/2026 [9], [8]	Real phantom	2.45 GHz ISM	4	random splits	3D / path RMSE	2.38 cm static, 0.38 cm path; 1.69 cm static, 0.39 cm path	real phantom ML point estimators; no posterior
Krishnan 2025 [21]	Sim	UWB 4.21–10.77 GHz	9–33	MC error injection + CRLB	3D RMSE	1.04 mm @33Rx	high-Rx UWB TDoA simulation; no posterior

B. Full-wave simulation

Channels are computed by full-wave electromagnetic simulation in CST Studio Suite 2025 using a full-wave sweep over 380 MHz to 430 MHz, centred on the legal MICS/MedRadio band 401 MHz to 406 MHz; the coherent sub-band is ~402 MHz to 405 MHz. The narrow-band production sweep agrees with the full-band reference to within 0.26 dB at 403 MHz. These are full-wave fields, not a parametric path-loss model, so the channel captures multipath and near-field coupling that bound-based prior art omits (Section II).

C. Dataset and feature representation

Each configuration is reduced to a 180-dimensional “rich” feature vector composed of five blocks: a 4-D received signal strength indicator (RSSI) block, 44-D multi-frequency magnitude, 44-D phase cosine, 44-D phase sine, and 44-D return loss. The combined dataset contains $N = 3935$ configurations spanning 788 unique capsule positions, with five orientation poses per position (identity, $z+90^\circ$, $y+90^\circ$, $y+180^\circ$, $x+90^\circ$). Two documented coverage gaps exist (a 15-configuration setup gap in one batch, and a final batch of 50 by design) and are accounted for in the analysis.

Combined set vs. initial-stage subset ($N = 1980$): The dense data were generated in two stages. We refer to the full set ($N = 3935$, all 788 positions) as the *combined set*, and to the initial-stage subset ($N = 1980$, the first 396 positions $\times$ 5 poses, generated before the full sweep completed) as the *initial-stage subset*. The headline point-accuracy and CFM posterior numbers use the combined set. The initial-stage subset has two uses: the multi-method UQ bake-off (Table VI) and the deployability ablation (Table VII), both under identical pipelines, geometry, and protocol. We tag every result with the set it uses. The same CFM model therefore reports a 0.354 cm point MAE on the combined set and 0.501 cm on the smaller subset, consistent with a data-size effect (same model, same pipeline).

D. Evaluation protocol

All methods are evaluated under **position-grouped 5-fold cross-validation** (GroupKFold over the 788 unique positions) with three seeds ($5 \times 3 = 15$ fold-seed evaluations). Grouping by position guarantees that all poses of a given position fall in the same fold, so no position is ever split across train and test; this is materially stricter than the random point-wise splits used by several comparators, and it removes

TABLE II
CST REPRODUCIBILITY CHECKLIST FOR THE CORRECTED
FOUR-RECEIVER SIMULATION DATA. **SIMULATION.**

Item	Repository-grounded value
Project and units	CST Studio 2025.0 Microwave Studio; length unit mm
Body dimensions	$273 \times 278 \times 380$ mm envelope in the paper coordinate convention ($x/y/z$, with y vertical); dense batch metadata records the same extents as $L_{\text{body}} = 273$, $W_{\text{body}} = 278$, and $H_{\text{body}} = 380$ in CST-native parameter names
Receiver locations	Processed dataset metadata uses four effective receiver centres: P1/P3 at $y = +67.5$ mm, P2/P4 at $y = -72.5$ mm; side-wall offsets are $z = \mp 190.835$ mm for P1/P2 and $x = \pm 137.335$ mm for P3/P4. Figure 2 is a schematic reconstruction, not a direct CST CAD export
Materials	Bio Tissue/Muscle body phantom, Bio Tissue/Skin outer layer, PVC body-wall layer, copper conductors, polyimide capsule substrate
Ports	Four external 50Ω S-parameter receiver ports on the $-z$, $+z$, $+x$, and $-x$ faces; capsule antenna port 6 on Antenn_in/Inner_antenna
Solver and boundaries	Frequency Domain (Tetrahedral); all six boundaries expanded open, with no symmetry planes
Mesh settings	Tetrahedral mesh entries with volume and surface gradation 1.5 and energy adaption are present in the replayed ModelHistory
Frequency sweep	Dense narrow-band S-parameter tensor over 380–430 MHz with 1001 frequency points
Generated artifacts	S-parameter batch tensors and a combined 180-D rich-feature dataset used for the $N = 3935$ analysis

TABLE III
CONSTRUCTION OF THE 180-D RICH RF FEATURE VECTOR. **SIMULATION.**

Block	Dimension	Definition
RSSI block	4	Columns 0–3: $ S_{i5} $ in dB at the bin nearest 400 MHz, one capsule-to-receiver value for each of the four receiver ports
Magnitude	44	Columns 4–48: $ S_{i5} $ in dB at 11 evenly spaced bins: 380, 385, 390, 395, 400, 405, 410, 415, 420, 425, 430 MHz
Phase cosine	44	Columns 48–92: $\cos(\arg S_{i5})$ at the same 11 frequency bins
Phase sine	44	Columns 92–136: $\sin(\arg S_{i5})$ at the same 11 frequency bins
Return loss	44	Columns 136–180: $ S_{ii} $ in dB at the same 11 frequency bins
Conventions	180 total	Magnitudes use $20 \log_{10} S $ with a -120 dB floor; phase is stored as cosine/sine to avoid wrapping; train-fold normalization uses training indices only

the position-leakage that inflates random-split scores. Normalization statistics are fit on the training fold only. To make method comparisons paired, the same test indices are reused across methods, and significance is assessed by a position-paired bootstrap over the 788 unique coordinates.

Caveat: The entire dataset is simulation. No measured-phantom or in-vivo data appear anywhere in this study.

IV. METHOD

A. Deterministic backbone (point reference)

As a point-accuracy reference we train a deterministic regression backbone on the rich feature vector with a mean-squared-error objective. The backbone is used *only* as the point-accuracy reference against which the CFM posterior’s point cost is measured; it produces no posterior and therefore

cannot be calibrated or ranked by confidence. Its accuracy is reported in Section V.

B. Conditional flow-matching posterior

The estimator of interest is a clean conditional flow-matching (CFM) model that learns a velocity field transporting a *cuboid-uniform bounded* base distribution, $x_0 \sim \text{Unif}(\mathcal{C})$ over the phantom cuboid $\mathcal{C}$, to the conditional distribution of capsule position given the observed features; sampling starts from the same bounded base and integrates the learned ODE with Euler updates, clamping every intermediate state to the cuboid bounds. We use the “clean” configuration: independent-coupling CFM over a shared vector-field backbone, exponential-moving-average weights with decay 0.999, and 200 posterior samples per test point. Minibatch OT-CFM was unstable on the small position-grouped batches, so it is not used for the reported posterior. Figure 3 summarizes the sampling pipeline.

C. Formal CFM objective and training specification

Let c denote the normalized RF feature vector and $x_1 \in \mathcal{C} \subset \mathbb{R}^3$ the capsule position. Training draws a bounded base point $x_0 \sim \text{Unif}(\mathcal{C})$ and $t \sim \text{Unif}(0, 1)$, then uses the straight-line path

$$x_t = (1 - t)x_0 + tx_1, \quad u_t = x_1 - x_0, \quad (1)$$

with the conditional flow-matching loss

$$\mathcal{L}(\theta) = \mathbb{E}_{(x_1, c), x_0, t} \left[\|v_\theta(x_t, t | c) - u_t\|_2^2 \right]. \quad (2)$$

The implementation adds a small $\sigma_{\min} = 0.001$ m target jitter before forming x_1 for training stability. At inference, each sample starts from the same cuboid-uniform base and is advanced by clamped Euler integration,

$$\hat{x}_{k+1} = \Pi_{\mathcal{C}}[\hat{x}_k + \Delta t v_\theta(\hat{x}_k, k\Delta t | c)], \quad \Delta t = 1/K, \quad (3)$$

where $\Pi_{\mathcal{C}}$ clips each coordinate to the cuboid bounds.

D. Posterior summaries and credible regions

From the posterior sample cloud we form the empirical posterior, report its mean as the point estimate, and construct credible regions at nominal 50/90/95% both as axis-aligned marginal boxes and as Mahalanobis ellipsoids. *Coverage* is the fraction of test points whose true position lies inside the nominal region; *discrimination* is the Spearman correlation between posterior spread and realized error. A selective-prediction (operating-point) view, in which the least-confident fraction of estimates is deferred, follows directly from the spread ordering. Figure 4 shows the empirical sample representation used for these summaries.

E. Metrics

For each held-out configuration $i = 1, \dots, N$, let x_i be the true 3D position, $\{s_i^{(m)}\}_{m=1}^M$ the posterior samples, and

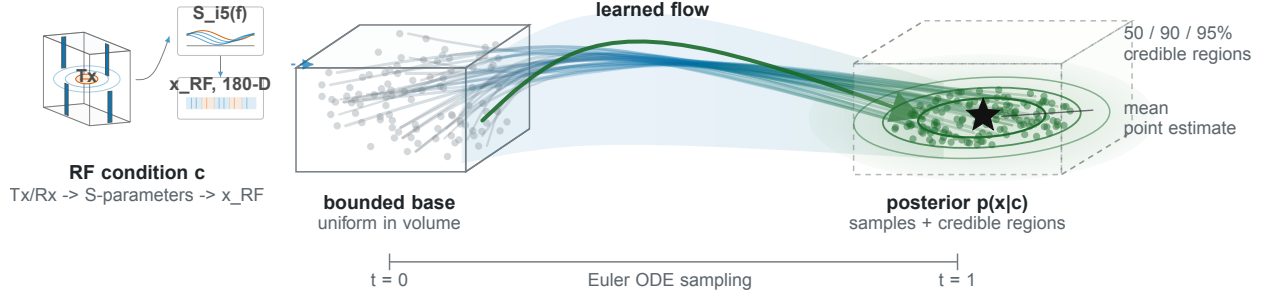


Fig. 3. Conditional flow-matching posterior pipeline. The 180-D RF feature vector conditions a velocity field $v_\theta(x, t | c)$ (FiLM blocks, EMA weights) that transports cuboid-uniform bounded base samples to the conditional position distribution; Euler ODE integration clamps each step to the cuboid bounds, and 200 draws form the empirical posterior, summarised by its mean (the point estimate) and 50/90/95% credible regions. **Simulation.**

TABLE IV

CFM ARCHITECTURE AND TRAINING HYPERPARAMETERS USED FOR THE REPORTED POSTERIOR. **SIMULATION.**

Item	Value
Dataset and folds	Combined rich dataset ($N = 3935$); seeds 1, 2, 3; 5 position-grouped folds
Epoch schedule	600 epochs, patience 40, validation every 5 epochs
Batch and optimizer	batch size 64; AdamW, learning rate 2×10^{-3} , weight decay 10^{-4} , betas (0.9, 0.95)
Base and path	$x_0 \sim \text{Unif}(\mathcal{C})$; uniform t ; linear path $x_t = (1-t)x_0 + tx_1$; target $x_1 - x_0$
Stabilization	$\sigma_{\min} = 0.001$ m target jitter; gradient clipping at 1.0
Coupling and weights	independent coupling (use_ot_cfm=false); EMA decay 0.999
Sampling	Euler integrator, $K = 50$ steps, 200 posterior samples per test point
Input split	leading 4-D RSSI token stream plus 176-D global rich-feature stream
Vector-field width	hidden dimension 64; 3 FiLM residual blocks
Attention and geometry	per-sensor geometry tokens with relative geometry; 4-head self-attention; learned-query pooling; dropout 0.1
Feature/time embeddings	PLR feature embedding $K = 24$, $\sigma = 0.1$, $d = 16$; Fourier time features $\sin / \cos(2\pi t, 4\pi t, 8\pi t, 16\pi t)$

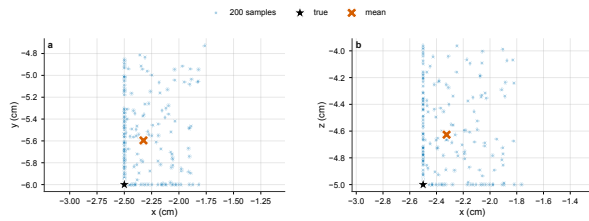


Fig. 4. Example CFM posterior cloud for one test point, shown in the x - y and x - z projections. The 200 samples define the empirical posterior; the posterior mean is the point estimate used for MAE, and the cloud spread defines the credible-region and selective-prediction scores. **Simulation.**

$\bar{x}_i = M^{-1} \sum_m s_i^{(m)}$ the posterior mean. The 3D point metric is

$$\text{MAE}_{3D} = \frac{1}{N} \sum_{i=1}^N \|\bar{x}_i - x_i\|_2, \quad (4)$$

reported in centimetres. For nominal levels $\alpha \in \{0.50, 0.90, 0.95\}$, let $q_{i,d}^-(\alpha)$ and $q_{i,d}^+(\alpha)$ be the

lower and upper sample quantiles on axis d . Marginal and joint-box coverage are

$$\text{cov}_{\text{marg}}(\alpha) = \frac{1}{3N} \sum_{i=1}^N \sum_{d=1}^3 \mathbf{1}\{q_{i,d}^-(\alpha) \leq x_{i,d} \leq q_{i,d}^+(\alpha)\}, \quad (5)$$

$$\text{cov}_{\text{joint}}(\alpha) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\forall d: q_{i,d}^-(\alpha) \leq x_{i,d} \leq q_{i,d}^+(\alpha)\}. \quad (6)$$

For a full-covariance Gaussian check, with local sample covariance Σ_i , Mahalanobis-ellipsoid coverage is

$$\text{cov}_{\text{ell}}(\alpha) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{(x_i - \bar{x}_i)^T \Sigma_i^{-1} (x_i - \bar{x}_i) \leq \chi_{3,\alpha}^2\}. \quad (7)$$

The sample CRPS is computed per axis and averaged,

$$\text{CRPS} = \frac{1}{3N} \sum_{i=1}^N \sum_{d=1}^3 \left[\frac{1}{M} \sum_{m=1}^M |s_{i,d}^{(m)} - x_{i,d}| - \frac{1}{2M^2} \sum_{m,\ell=1}^M |s_{i,d}^{(m)} - s_{i,d}^{(\ell)}| \right], \quad (8)$$

and the multivariate energy score is

$$\text{ES} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{M} \sum_{m=1}^M \|s_i^{(m)} - x_i\|_2 - \frac{1}{2M(M-1)} \sum_{m \neq \ell} \|s_i^{(m)} - s_i^{(\ell)}\|_2 \right]. \quad (9)$$

Sharpness is the mean posterior standard deviation over test points and axes,

$$\text{sharpness} = \frac{1}{3N} \sum_{i=1}^N \sum_{d=1}^3 \text{sd}_m(s_{i,d}^{(m)}). \quad (10)$$

F. Orientation as a marginalized nuisance

Capsule orientation in vivo is unknown and time-varying. We therefore treat orientation as a nuisance variable to be *integrated out*, not regressed. Two mechanisms accomplish this: (i) the backbone is approximately orientation-invariant

across the sampled poses, and (ii) the CFM posterior’s *width* absorbs the residual orientation-induced ambiguity. Section V quantifies the per-pose bias, the 180° y -flip check, and how faithfully the posterior width tracks the orientation-induced spread of the point estimate.

G. Baselines

Against the CFM posterior we evaluate: the deterministic backbone (point reference); a 5-member deep ensemble; MC-dropout (50 samples); diagonal and 2-component mixture-density networks (MDN-dia, MDN-mix2; 200 samples); split conformal applied to the deterministic backbone; and ten classical regressors (HistGradientBoosting, ExtraTrees, RandomForest, SVR, NGBoost, k -NN, MLP, AdaBoost, Ridge, and a fixed kernel Gaussian process). Every method is evaluated on identical folds, seeds, and test indices.

V. RESULTS

All numbers are full-wave CST simulation under position-grouped 5-fold CV. The headline CFM posterior and point numbers are on the combined $N = 3935$ set; the multi-method uncertainty comparison (Table VI) is on the initial-stage subset ($N = 1980$) and is labelled as such. The headline result is calibration with discrimination: the CFM posterior is the only evaluated method that clears point error, multi-level coverage, and error-spread ranking together, while preserving backbone-level point accuracy within seed variation. We report the point-accuracy reference first (Section V-A), then the calibration result (Section V-B), the deployability ablation (Section V-C), a controlled observability probe of posterior modality (Section V-D), and the orientation-marginal width result (Section V-E).

A. Point accuracy: the posterior preserves backbone accuracy within seed variation

Table V supports the point-cost claim on the combined $N = 3935$ rich set. The clean CFM posterior (0.354 cm, 95% CI [0.339, 0.369]) ties the deterministic backbone (0.340(4) cm on the combined $N = 3935$ set; 0.340 cm in Table V) within seed noise and significantly outperforms all ten classical regressors: every position-paired 95% bootstrap interval on the difference excludes zero, with the best classical method (HistGradientBoosting, 0.779 cm) trailing CFM by +0.424 cm [0.405, 0.443]. The deterministic backbone’s per-axis errors are $x/y/z = 0.14/0.17/0.18$ cm; the per-axis CFM error is balanced ($x/y/z = 0.143/0.181/0.185$ cm). On the corrected geometry the y axis is no longer the entire error budget. The CFM-vs-backbone tie lacks a paired CI because the backbone’s per-point predictions were not retained; we therefore report it as “within seed noise,” not as a significance test. Figure 5 shows that CFM preserves backbone-level point accuracy while beating the classical regressors, Figure 6 confirms the corrected geometry balances the axes, and Figure 7 shows that the residual error is concentrated on the no-face-coverage y axis.

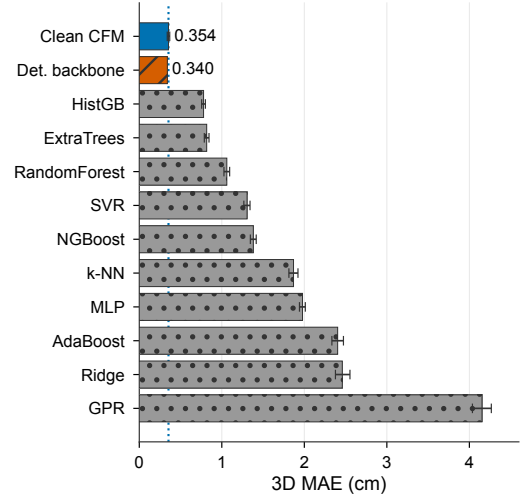


Fig. 5. Point-accuracy ranking on the combined rich set ($N = 3935$). Horizontal bars show 3D MAE; error bars show position-paired 95% bootstrap CIs where available. Clean CFM ties the deterministic backbone within seed noise and outperforms the classical regressors under identical position-grouped folds. **Simulation.**

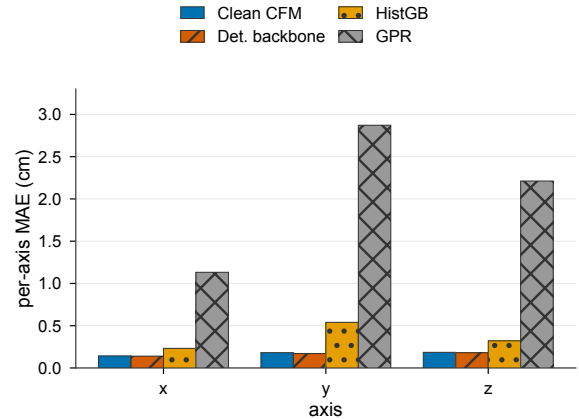


Fig. 6. Per-axis MAE on the combined rich set ($N = 3935$). The anti-symmetric receiver placement leaves y as the hardest axis but breaks the mirror symmetry enough for CFM and the deterministic backbone to remain balanced; GPR is dominated by y error. **Simulation.**

B. Calibration: the only method clearing all three bars

Table VI is the central result. In the multi-method comparison on the initial-stage subset ($N = 1980$), the clean CFM posterior is the *only* method that simultaneously achieves competitive point error, coverage from its own samples that is *conservative* at the 50% level and near-nominal at 90/95% (marginal 50/90/95% = 0.766/0.935/0.944), and real error-spread discrimination ($\rho = 0.601$). The deep ensemble attains marginally lower MAE (0.359 cm) and higher ρ (0.705) but *under-covers* badly ($\text{cov}_{90} = 0.717$); MC-dropout under-covers severely ($\text{cov}_{90} = 0.464$). Both mixture-density networks *over-cover* ($\text{cov}_{90} \approx 0.96$ – 0.97) with near-zero discrimination ($\rho \leq 0.10$): they widen without ranking. Split conformal targets nominal coverage; its observed $\text{cov}_{90} = 0.893$ and it likewise carries $\rho \approx 0$. On the full combined

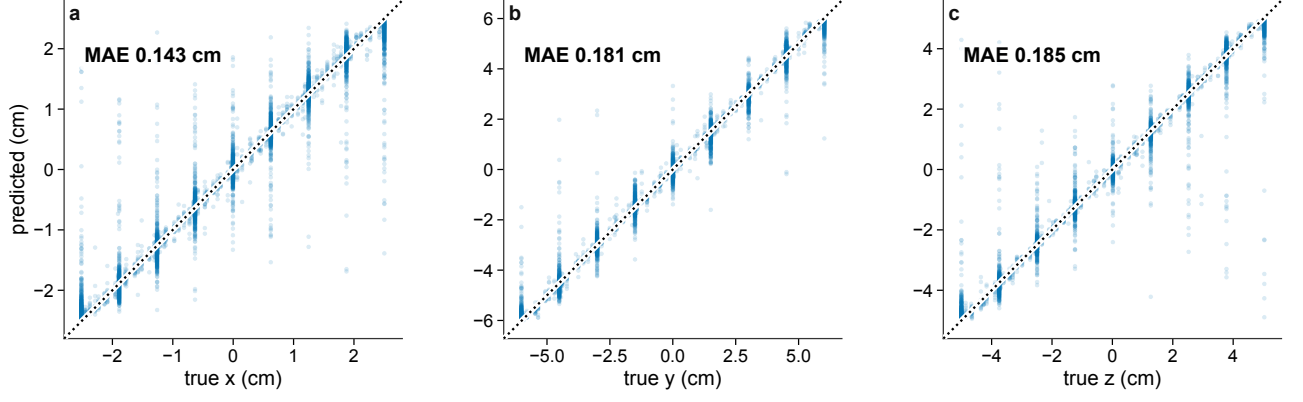


Fig. 7. CFM predicted versus true position per axis on the combined set ($N = 3935$); points on the dotted identity line are exact. The x and z axes are tightly recovered, while y —the axis with no face receivers—carries most of the residual error, consistent with Fig. 6. **Simulation.**

TABLE V
POINT ACCURACY ON THE COMBINED RICH SET ($N = 3935$,
POSITION-GROUPED 5-FOLD CV, 3 SEEDS), 3D MAE WITH
POSITION-PAIRED 95% BOOTSTRAP CIs (PAIRED VS. CFM OVER THE 788
UNIQUE POSITIONS). **SIMULATION.**

Method	3D MAE (cm)	Paired Δ vs. CFM (cm)
Deterministic backbone (point ref.)	0.340	tie [†]
Clean CFM (posterior mean)	0.354 [0.339, 0.369]	—
HistGradientBoosting	0.779 [0.756, 0.801]	+0.424 [0.405, 0.443]
ExtraTrees	0.817 [0.788, 0.845]	+0.462 [0.439, 0.486]
RandomForest	1.061 [1.026, 1.095]	+0.706 [0.677, 0.735]
SVR	1.306 [1.267, 1.342]	+0.952 [0.920, 0.983]
NGBoost	1.382 [1.346, 1.417]	+1.028 [0.994, 1.060]
k -Neighbors	1.869 [1.813, 1.923]	+1.515 [1.462, 1.565]
MLPRegressor	1.978 [1.941, 2.012]	+1.624 [1.592, 1.652]
AdaBoost	2.404 [2.334, 2.474]	+2.052 [1.984, 2.116]
Ridge	2.460 [2.377, 2.553]	+2.108 [2.026, 2.194]
GPR (fixed kernel)	4.153 [4.037, 4.266]	+3.800 [3.687, 3.908]

[†]Tie within seed noise; paired CI not available because the backbone per-point predictions were not retained for this analysis. All values are full-wave CST simulation under position-grouped 5-fold CV.

$N = 3935$ set the CFM posterior is consistent and slightly conservative (marginal 50/90/95% = 0.832/0.970/0.975; joint 0.624/0.925/0.939; pooled $\rho = 0.58$, fold-mean 0.62; CRPS 0.127 cm, energy score 0.277 cm, sharpness 0.281 cm; Fig. 10). The deterministic backbone produces no posterior. Of the classical regressors, only NGBoost and the fixed-kernel Gaussian process emit a predictive variance, but both are *parametric and unimodal*—they cannot represent the multi-modality that is the point of Section V-D—and their point error (Table V) is far from the operating region of interest; we therefore restrict the uncertainty panel to the sampling-based posteriors and the conformal baseline. Figure 8 shows that only CFM occupies the near-nominal, high-discrimination region, Figure 9 confirms the multi-level coverage pattern, and Figure 11 shows that CFM spread ranks realized error on the combined set. Because the spread ranks error, deferring the least-confident estimates sharply reduces the error of the retained set—at 50% retention the 3D MAE falls from 0.354 cm to ~ 0.18 cm, tracking the oracle ordering (Fig. 12)—

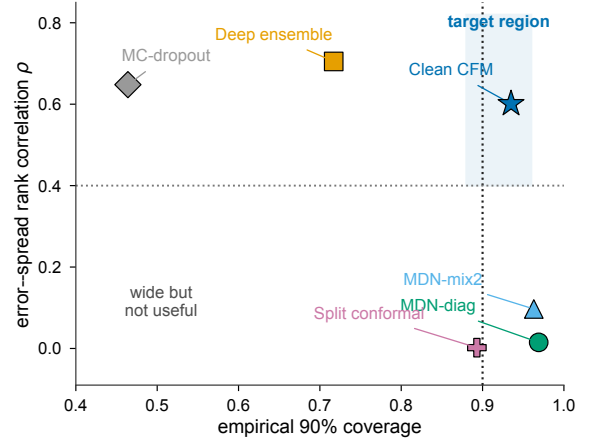


Fig. 8. Calibration versus discrimination in the initial-stage subset ($N = 1980$). The horizontal axis is empirical 90% coverage and the vertical axis is Spearman error-spread correlation; CFM is the only method in the near-nominal, high-discrimination region. **Simulation.**

which is the basis for a selective-prediction operating point.

C. Deployability ablation: RF phase, not bandwidth, is the lever

A single-tone magnitude-only MICS/MedRadio device cannot measure phase, so Table VII and Figure 13 ablate which feature components drive sub-cm accuracy on the initial-stage subset ($N = 1980$). This single-tone magnitude-only ablation is the strict-legal-band evidence. The *single-tone, magnitude-only* floor (S1) is ~ 1.5 cm (CFM 1.519 cm, HistGB 1.550 cm, ExtraTrees 1.731 cm). Adding phase at a single tone (S2) drops HistGB to 0.970 cm (-37%) and CFM to 0.524 cm; recovering phase across the coherent sub-band ~ 402 MHz to 405 MHz (S3) reaches CFM 0.481 cm, matching the full rich representation on the same subset (S5, CFM 0.512 cm). Crucially, *multi-frequency magnitude alone* (S4) does not

TABLE VI
UQ BAKE-OFF ON THE INITIAL-STAGE SUBSET ($N = 1980$), MARGINAL COVERAGE. **BOLD** = THE ONLY METHOD CLEARING POINT + NEAR-NOMINAL COVERAGE + DISCRIMINATION TOGETHER. **SIMULATION.**

Method	MAE (cm)	cov ₅₀	cov ₉₀	cov ₉₅	$\rho_{\text{err,spread}}$
Clean CFM	0.501	0.766	0.935	0.944	0.601
Deep ensemble (5)	0.359	0.402	0.717	0.738	0.705
MC-dropout (50)	0.473	0.206	0.464	0.523	0.648
MDN-diag (200)	0.533	0.874	0.969	0.978	0.015
MDN-mix2 (200)	0.614	0.837	0.963	0.973	0.097
Split conformal	—	0.483	0.893	0.945	0.002

All values are full-wave CST simulation under position-grouped 5-fold CV. Split conformal wraps the deterministic backbone's point estimate; its point MAE is therefore not repeated. Combined-set CFM-only values are reported in the text.

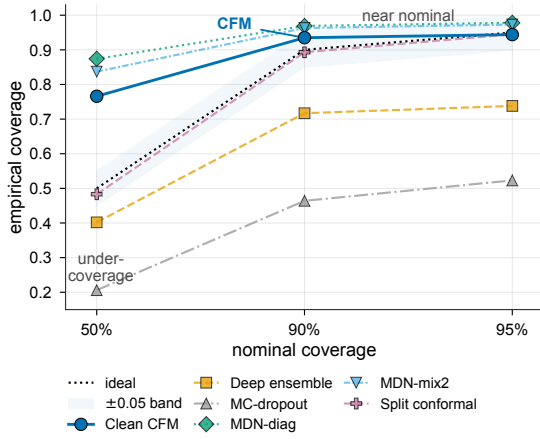


Fig. 9. Coverage reliability on the initial-stage subset ($N = 1980$), marginal. Clean CFM is conservative at the 50% level and close to the diagonal at 90/95%; the deep ensemble and MC-dropout under-cover; the MDNs over-cover. Combined-set CFM is slightly conservative (text). **SIMULATION.**

reach the sub-centimetre regime (CFM 1.191 cm, HistGB 1.652 cm): bandwidth without phase coherence is not enough. The deployable magnitude-only number to quote is ~ 1.5 cm; sub-centimetre accuracy requires phase-coherent receivers. Because this ablation uses the smaller initial-stage subset ($N = 1980$), the S5 CFM value should not be compared directly with the combined-set CFM headline in Table V.

D. Geometry ablation: receiver placement sets localizability, and the posterior reports it

Localization quality is ultimately set by *where the receivers are*: a four-receiver array makes a coordinate recoverable only if the receivers break the symmetry along that axis. We make this concrete with a controlled receiver-geometry ablation and ask not only “how accurate?” but “does the estimator report when an axis is unrecoverable?” We compare two arrays (Fig. 15): a *mirror-symmetric* array with all four receivers in the $y = 0$ plane (so the forward operator is invariant to $y \rightarrow -y$ and the y coordinate is unobservable by construction), and the main anti-symmetric side-wall array, whose $\pm x/\pm z$ wall placement breaks that symmetry and recovers y . This is

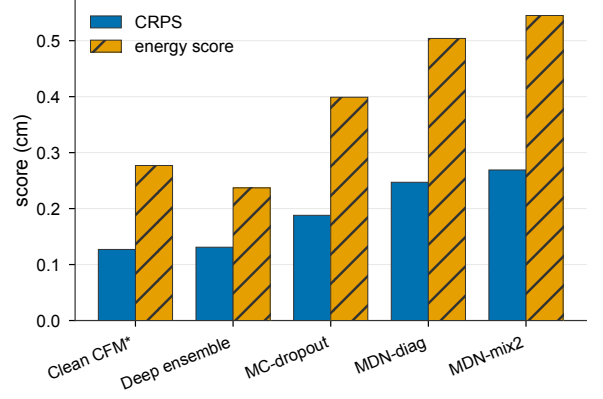


Fig. 10. Proper scoring rules (lower is better): CRPS and energy score per method. CFM* is scored on the combined set while the other methods use the initial-stage subset ($N = 1980$, as in Table VI), so the bars are *not* directly comparable; on its own set CFM attains a low CRPS and a competitive energy score. **SIMULATION.**

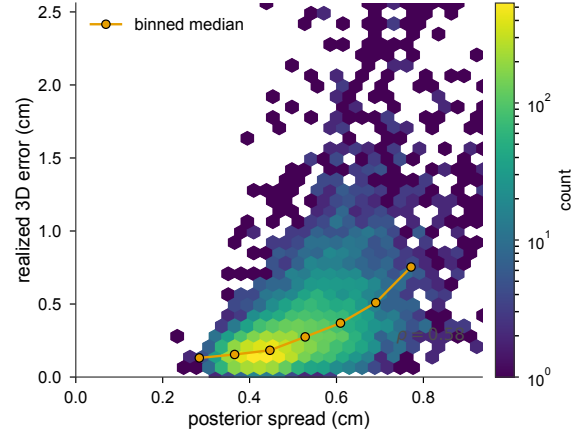


Fig. 11. Per-point CFM error versus posterior spread on the combined set ($N = 3935$). Each point is a held-out configuration; the positive rank correlation shows that the posterior width is useful for selective prediction, not merely wider on average. **SIMULATION.**

why geometry matters in localization, and it is the regime in which a *point* estimator is misleading: when two positions (a coordinate and its mirror) explain the measurement equally well, a single point silently picks one.

A generative posterior, by contrast, can represent *both* hypotheses. On the mirror-symmetric array the CFM posterior is genuinely *bimodal* (Fig. 14): it captures the true position’s mirror partner in 49.7% of high- $|y|$ cases and attains joint 90% coverage 0.946, while the 3D point error sits at the ~ 2.5 cm unobservable- y floor—the posterior mean falls *between* the two modes, which is exactly why a single point estimate is misleading here. The competing probabilistic methods fail to represent the ambiguity in opposite ways: a 2-component mixture-density network *covers but does not split* (it widens into one blob), and the deep ensemble *collapses* (joint 90% coverage 0.017, zero mirror capture). On the anti-symmetric side-wall array the same CFM posterior

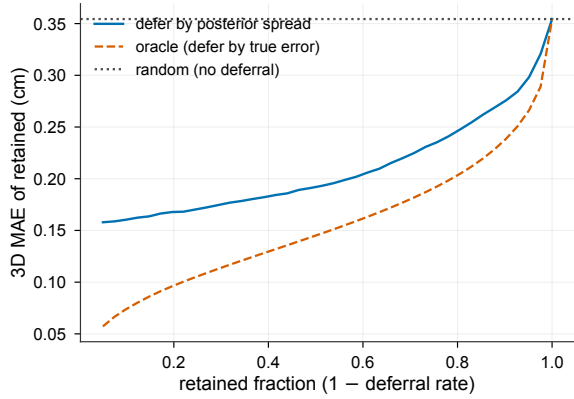


Fig. 12. Selective prediction on the combined set ($N = 3935$). Deferring the least-confident estimates by posterior spread (solid) monotonically lowers the 3D MAE of the retained set and closely tracks the oracle ordering by true error (dashed); the dotted line is the no-deferral mean. The gap between the spread and oracle curves is the residual room for a better confidence signal. **Simulation.**

TABLE VII

FEATURE-DEPLOYABILITY ABLATION ON THE INITIAL-STAGE SUBSET ($N = 1980$) (3D MAE, CM). PHASE COHERENCE (S2/S3/S5), NOT BANDWIDTH ALONE (S4), SEPARATES THE ~ 1.5 cm SINGLE-TONE FLOOR FROM THE SUB-CM REGIME. **SIMULATION.**

Feature set	HistGB	ExtraTrees	CFM
S1 Single-tone magnitude only	1.550	1.731	1.519
S2 + phase at one tone	0.970	1.239	0.524
S3 Coherent sub-band $\sim 402\text{--}405$ MHz	1.063	1.140	0.481
S4 Multi-frequency magnitude only	1.652	1.688	1.191
S5 Full rich (180-D)	1.050	1.146	0.512

All values are full-wave CST simulation under position-grouped 5-fold CV on the initial-stage subset ($N = 1980$).

is *unimodal*, because the geometry makes y observable in simulation. Posterior *modality* therefore tracks the invertibility of the receiver geometry within this controlled ablation, and only the generative flow follows the modality change faithfully. The mirror-symmetric array is a deliberate ablation geometry, distinct from the main side-wall dataset; as Table VIII makes explicit, the two arrays also differ in feature representation (4-D magnitude-only RSSI for the symmetric array versus the 180-D rich vector for the main array), so this is not a feature-matched comparison. The bimodality is nonetheless attributable to geometry rather than feature richness: the mirror-symmetric forward map is invariant to $y \rightarrow -y$ by construction, so y is unobservable for any feature set, and a richer feature vector cannot remove an exact geometric symmetry.

E. Posterior width carries orientation uncertainty

On the main dataset, changing only the capsule’s orientation at a fixed position moves the RF fingerprint more than physically relocating the capsule does (within-position-across-pose spread ratio $\approx 1.17\text{--}1.22$). The backbone absorbs this by invariance: the per-pose systematic bias is tiny (weighted-RMS $x/y/z = 0.027/0.005/0.046$ cm, all below 0.05 cm per axis; maximum signed bias 0.074 cm), and a 180° flip about y does not flip the y estimate. The orientation-induced spread

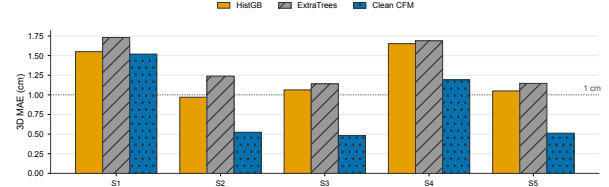


Fig. 13. Grouped-bar view of the feature-deployability ablation. Phase information at one tone or across the coherent sub-band $\sim 402\text{--}405$ MHz moves CFM into the sub-cm regime and gives the tree models a clear phase benefit (ExtraTrees and HistGB improve, though not all reach sub-cm); additional magnitude-only bandwidth does not. **Simulation.**

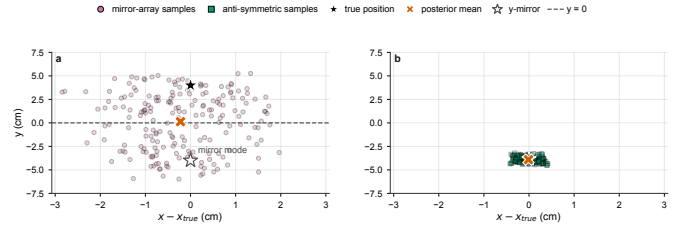


Fig. 14. Posterior modality tracks receiver-geometry invertibility, shown with real CFM samples for a high- $|y|$ test point (the x - y plane). *Left*: on the mirror-symmetric array the posterior is *bimodal*—it places mass on both the true position and its y -mirror, so the posterior mean falls between the modes and the point estimate is misleading. *Right*: on the anti-symmetric side-wall array the posterior for a comparable high- $|y|$ point is *unimodal* and concentrated on the truth (note the shared y -scale). **Simulation.**

of the point estimate is small in the mean ($x/y/z/3D = 0.123/0.145/0.187/0.320$ cm) but has a heavy tail (max 3D 3.25 cm). Critically, the posterior *width* tracks that orientation-induced spread at Spearman $\rho = 0.748/0.826/0.828$ ($x/y/z$; 3D 0.847), and the orientation-marginal posterior (pooling a position’s poses) is $1.20\text{--}1.36\times$ wider than the pose-conditioned posterior while covering 100% of the per-pose cloud—a conservative representation of unknown pose. As a diagnostic, pose is a five-way classification problem (random-forest accuracy $74.9\% \pm 2.5\%$ vs 20% chance): the out-of-plane $z90/x90$ poses are perfectly separable (recall 1.00) while the in-plane y -family {identity, $y90$, $y180$ } is degenerate (recall 0.55–0.61)—the orientation-space mirror of the y -position blind spot. Only five axis-aligned poses are sampled, so this “orientation-marginal” is a five-sample approximation, not a true $SO(3)$ marginal (Section VI). Figure 16 confirms that marginalizing unknown pose widens the posterior while conservatively covering the fixed-position pose cloud.

VI. DISCUSSION

A. The contribution is a phenomenon plus a faithful representation

CFM is useful here because it preserves point accuracy while representing two forms of uncertainty that point estimators hide: calibration error and geometry-induced multimodality. First, calibration is not free: among methods that produce a posterior, only the clean CFM is simultaneously accurate, well covered from its own samples (conservative at 50%, near-nominal at 90/95%), and discriminative (Table VI); ensembles

TABLE VIII

SELF-CONTAINED RECEIVER-ARRAY MODALITY ABLATION. THE MIRROR-SYMMETRIC ROW TESTS AN INTENTIONALLY NON-INVERTIBLE y GEOMETRY; THE ANTI-SYMMETRIC SIDE-WALL ROW IS THE CORRECTED-GEOMETRY INITIAL-STAGE SUBSET RUN USED TO ESTABLISH THE UNIMODAL COMPARATOR. **SIMULATION.**

Array	Dataset and feature dimension	CFM configuration	Samples	Mirror capture (%)	Joint 90% coverage	3D point error (cm)
Mirror-symmetric array	4-D RSSI ablation dataset; all sensor y values are 0.0	no-OT, no-EMA, Euler; 600 epochs	300	49.7	0.946	2.546
Anti-symmetric side-wall array	Initial-stage subset ($N = 1980$); 180-D rich features	no-OT, EMA 0.999, Euler; 600 epochs	200	0.0	0.856	0.501

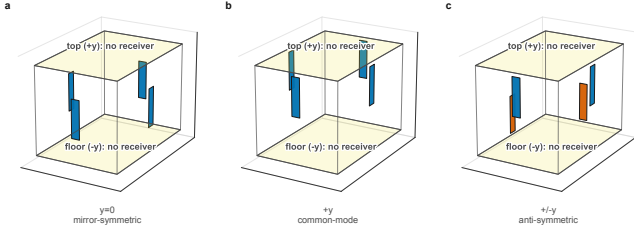


Fig. 15. Receiver-placement ablation, drawn with y (height) vertical; the shaded $\pm y$ top/floor faces carry no receiver and only the receiver height (y -coordinate) changes between panels. (a) Mirror-symmetric array with all four receivers at the same height ($y = 0$): the forward map is invariant to $y \rightarrow -y$, so y is unobservable and the CFM posterior is bimodal. (b) Common-mode shift ($+y$): weak, biased y information. (c) The main side-wall array places the receivers anti-symmetrically in height (green upper, orange lower), breaking the symmetry so that y becomes observable and the posterior is unimodal. **Simulation.**

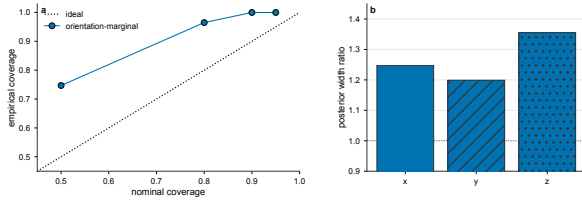


Fig. 16. Orientation-marginal uncertainty on the main dataset. The left panel shows empirical coverage of the fixed-position pose cloud by the orientation-marginal posterior; the right panel shows that marginalizing unknown pose widens the posterior relative to pose-conditioned sampling. **Simulation.**

under-cover and mixture-density networks over-cover without ranking. Second, posterior *modality* tracks forward-operator invertibility (Section V-D): in the mirror-symmetric ablation the posterior is bimodal when y is unobservable, while on the anti-symmetric side-wall array it is unimodal, and only the generative flow follows that modality change. A deterministic regressor and its implied point estimate cannot represent this at all; and in our ablation the tested deep ensemble collapsed while the 2-component mixture-density network widened into a single blob rather than splitting, whereas the flow represents both hypotheses while preserving point accuracy within 0.014 cm of the backbone (0.354 cm vs 0.340 cm).

B. Why calibration, not point accuracy, is the right target here

At the sub-centimetre level, the marginal point-accuracy differences between the best methods are small relative to the simulation-to-reality gap, whereas a calibrated, discriminative posterior could support clinically relevant selective-prediction

workflows after phantom and in-vivo validation: it can defer or re-measure the least-confident estimates and state where the capsule is *not*. This is why we pair the point-accuracy comparison of Table I with the uncertainty comparison of Table VI rather than reporting either alone.

C. What scales with data, and what does not

On this geometry, point error is largely capped by observability, while uncertainty quality continues to improve with data—the calibration and ranking metrics strengthen with dataset expansion even where the point error barely moves. Future work should recompute data-scaling and capacity sweeps on the corrected side-wall geometry before using them to make quantitative claims.

D. Limitations and the next evidence gate

- **Simulation only.** No measured-phantom or in-vivo data appear in this study. Real-data calibration—does the posterior remain calibrated on measured channels?—is the explicit next gate, and a physical-phantom companion is the single highest-value follow-up.
- **No y -face receivers.** The four-receiver vertical-wall geometry leaves the y axis observability-limited; the corrected geometry resolves it in simulation, but the hardware asymmetry is real.
- **Phase requires coherent receivers.** The sub-centimetre rich regime is not available to a simple single-tone magnitude-only MICS/MedRadio device; the deployable magnitude-only floor is ~ 1.5 cm (Table VII).
- **Five poses, not $SO(3)$.** Orientation invariance and the orientation-marginal posterior are demonstrated on five axis-aligned poses only; an arbitrary tilt is untested and the marginal is a five-sample approximation. A denser-orientation sweep is specified as future work.
- **Secondary statistics.** The CFM-vs-backbone comparison is reported as a tie within seed noise because paired deterministic-backbone predictions were not retained. No clinician feedback has yet informed a selective-prediction operating threshold.

VII. CONCLUSION

We presented, in full-wave simulation, the calibrated posterior: a flow-matching/generative WCE-RF position posterior with empirical multi-level coverage under position-grouped cross-validation at a four-receiver MICS/MedRadio budget in the legal MICS/MedRadio band 401 MHz to 406 MHz. It is

the only evaluated method that is simultaneously accurate, well covered from its own samples (conservative at 50%, near-nominal at 90/95%), and discriminative in error-vs-spread. Its posterior mean also preserves deterministic-backbone point accuracy within 0.014 cm (0.354 cm vs 0.340 cm) and beats ten classical regressors on identical position-grouped folds. Beyond aggregate calibration, we showed two properties unique to a generative posterior: in a controlled mirror-symmetric receiver-array ablation its *modality* tracks forward-operator invertibility, and on the main dataset its *width* carries the aleatoric uncertainty of an unknown capsule orientation. A feature-deployability ablation places the single-tone magnitude-only floor at ~ 1.5 cm and attributes the sub-centimetre regime to phase coherence rather than bandwidth.

We make no claim to state-of-the-art point accuracy; real-phantom systems at higher frequencies are more accurate, and we declare the simulation-vs-real and band confounds throughout. To our knowledge, the novelty is the first flow-matching/generative WCE-RF position posterior with empirical multi-level coverage under position-grouped cross-validation. The next evidence gate is a physical-phantom companion that tests whether the posterior remains calibrated on *measured* channels under the same position-grouped protocol. We also flag a true $SO(3)$ orientation marginal and paired deterministic-backbone predictions for a direct CFM-vs-backbone interval as concrete next steps.

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